QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBTG303 – MARINE BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (

Students should attempt at least one question from each course outcome to enhance their overall outcome attainabi

Part A

One Word Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Name the major pollutant responsible for marine plastic pollution. [An] / [CO4]
2. List the major levels of marine biodiversity. [R] / [CO1]
3. Which sampling method is used to collect organisms living on the ocean floor? [U] / [CO3]
4. List any two applications of marine biotechnology. [K] / [CO1]
5. Recall the name of the molecule in microalgae mainly used for biodiesel production. [An] / [CO2]
6. Which term refers to the establishment of protected areas for conserving marine species? [An] / [CO4]
7. What is called for the organisms adapted to high salt concentrations? [R] / [CO1]

8.	Name any two microorganisms commonly found in extreme marine environments.	[R] / [CO2]
9.	Which method is used to determine the concentration of microorganisms in a marine sample?	[U] / [CO3]
10.	Mention one importance of genetic diversity in marine microorganisms.	[R] / [CO1]
11.	What is CTD?	[U] / [CO3]
12.	Name the marine photosynthetic organisms which form the base of many marine food chains.	[R] / [CO1]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13.	Explain the methods used for collecting marine microorganisms.	[U] / [CO3]
14.	Compare insitu and exsitu conservation strategies for marine organisms.	[An] / [CO4]
15.	Define marine biotechnology and state its scope by listing the major group of marine organisms.	[R] / [CO1]
16.	Differentiate between plankton net and a sediment corer in sampling.	[U] / [CO3]
17.	Analyse the major challenges associated with marine drug development.	[An] / [CO4]
18.	Compare microbial diversity in different marine habitats and analyse the factors responsible for it.	[R] / [CO2]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

19.	Explain the principle of DNA sequencing and its importance and applications in marine biotechnology.	[U] / [CO3]
20.	Compare the opportunities and challenges in utilizing marine biological resources.	[An] / [CO4]
21.	Examine the relationship between marine microbial diversity and the discovery of novel biotechnological products.	[R] / [CO2]
22.	Describe the techniques of sampling and role of analytical techniques in marine biology.	[U] / [CO3]

[15x2 = 30]